LAB ASSIGNMENT – IV

1. Create a VIEW that displays all buses starting from a source name starting with "ISBT".

```
himansh1241 — mysql -u root — 83×5

mysql> CREATE VIEW ISBT_Buses AS
    -> SELECT *FROM Buses
    -> WHERE SOURCE LIKE 'ISBT%';
Query OK, 0 rows affected (0.005 sec)
```

2. Display source locations that have more than 2 buses originating from them.

```
himansh1241 — mysql -u root — 83×12

mysql> SELECT Source, COUNT(Bus_no)
    -> FROM Buses
    -> GROUP BY Source
    -> HAVING COUNT(*) > 2;
+------------+----------------+
| Source     | COUNT(Bus_no)  |
+------------+----------------+
| Pari Chauk |              3 |
| ISBT       |              4 |
+------------+----------------+
2 rows in set (0.001 sec)
```

3. Show bus records that are scheduled to depart after September 25, 2022, ordered by departure date and time.

```
himansh1241 — mysql -u root — 83×13

mysql> SELECT *FROM Buses
    -> WHERE Date > '2022-09-25'
    -> ORDER BY Date, Time;
+--------+-----------+------------+-------------+------+------------+----------+
| Bus_no | Bus_name  | Source     | Destination | Fare | Date       | Time     |
+--------+-----------+------------+-------------+------+------------+----------+
| UP505  | Goluxury  | ISBT       | Kanpur      | 2250 | 2022-09-27 | 19:38:00 |
| DL701  | Vaishnavi | ISBT       | Chandigarh  | 1550 | 2022-09-28 | 23:55:00 |
| DL306  | Shatabdi  | ISBT       | Dehradun    | 1007 | 2022-09-29 | 20:45:00 |
| UP501  | Safar     | Pari Chauk | Varanasi    | 1080 | 2022-09-30 | 08:35:00 |
+--------+-----------+------------+-------------+------+------------+----------+
4 rows in set (0.001 sec)
```

4. Create a view named "Confirmed_Passengers" that displays details of passengers with a confirmed booking status.

```
himansh1241 — mysql -u root — 83×5
mysql> CREATE VIEW Confirmed_Passengers AS
    -> SELECT *FROM Passengers
    -> WHERE Status = 'Confirm';
Query OK, 0 rows affected (0.004 sec)
```

5. Retrieve the details of confirmed passengers from the "Confirmed_Passengers" view, ordered by age in descending order, and then by passenger names.

```
himansh1241 — mysql -u root — 83×15
mysql> SELECT *FROM Confirmed_Passengers
    -> ORDER BY Age DESC, P_name;
+--------------+----------+--------+------+--------+---------+
| Passenger_id | P_Name   | Bus_no | Age  | Gender | Status  |
+--------------+----------+--------+------+--------+---------+
| SCS015       | Abhishek | UP505  |   57 | M      | Confirm |
| SCS030       | Himanshu | UP501  |   23 | M      | Confirm |
| SCS056       | Satya    | UP101  |   23 | M      | Confirm |
| SCS043       | Rupam    | UP501  |   22 | F      | Confirm |
| SCS071       | Rahul    | DL306  |   21 | M      | Confirm |
| SCS014       | Diya     | HR303  |   19 | F      | Confirm |
| SCS012       | Arjun    | UP301  |   17 | M      | Confirm |
| SCS013       | Anamika  | HR302  |   15 | F      | Confirm |
+--------------+----------+--------+------+--------+---------+
8 rows in set (0.001 sec)
```

6. Retrieve the total count of passengers in each status category, ordered by count in descending order.

```
himansh1241 — mysql -u root — 83×11
mysql> SELECT Status, COUNT(P_Name)
    -> FROM Passengers
    -> GROUP BY Status
    -> ORDER BY COUNT(P_Name) DESC;
+---------+---------------+
| Status  | COUNT(P_Name) |
+---------+---------------+
| Confirm |             8 |
| Waiting |             4 |
+---------+---------------+
2 rows in set (0.001 sec)
```

7. Create a view that lists buses that are fully booked.

```
himansh1241 — mysql -u root — 83×5
mysql> CREATE VIEW Fully_Booked_Buses AS
    -> SELECT *FROM Bus_Status
    -> WHERE Available_seats = 0;
Query OK, 0 rows affected (0.002 sec)
```

8. Create a view that displays buses with low available seats (where available seats are less than 10).

```
himansh1241 — mysql -u root — 83×5
mysql> CREATE VIEW Low_Seat_Buses AS
    -> SELECT *FROM Bus_Status
    -> WHERE Available_seats < 10;
Query OK, 0 rows affected (0.002 sec)
```

9. Retrieve a list of buses ordered by the number of booked seats in descending order.

```
himansh1241 — mysql -u root — 83×16
mysql> SELECT *FROM Bus_Status
    -> ORDER BY Booked_seats DESC;
+--------+-----------------+--------------+
| Bus_no | Available_seats | Booked_seats |
+--------+-----------------+--------------+
| UP505  |               1 |           59 |
| UP101  |              10 |           51 |
| HR303  |              14 |           48 |
| DL701  |               8 |           46 |
| DL306  |              17 |           45 |
| UP301  |              35 |           25 |
| HR302  |              41 |           24 |
| UP501  |              50 |            0 |
+--------+-----------------+--------------+
8 rows in set (0.001 sec)
```

10. Retrieve a list of buses ordered by the total number of seats (available seats + booked seats) in ascending order.

himansh1241 — mysql -u root — 83×20

```
mysql> SELECT Bus_no,
    ->        Available_seats,
    ->        Booked_seats,
    ->        (Available_seats + Booked_seats) AS Total_Seats
    -> FROM Bus_Status
    -> ORDER BY Total_Seats ASC;
+--------+-----------------+--------------+-------------+
| Bus_no | Available_seats | Booked_seats | Total_Seats |
+--------+-----------------+--------------+-------------+
| UP501  |              50 |            0 |          50 |
| DL701  |               8 |           46 |          54 |
| UP301  |              35 |           25 |          60 |
| UP505  |               1 |           59 |          60 |
| UP101  |              10 |           51 |          61 |
| HR303  |              14 |           48 |          62 |
| DL306  |              17 |           45 |          62 |
| HR302  |              41 |           24 |          65 |
+--------+-----------------+--------------+-------------+
8 rows in set (0.000 sec)
```